Optimization for Deep Learning (OPT4DL)

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Gradient Descent: Performance under Convexity

Gradient Descent (GD): A Basic Optimization Method



- Consider

$$\min_{w \in \mathbb{R}^d} f(w) \tag{1}$$

- We talked about a simple method based on Fixed Point Calculation called Gradient Descent (GD)
  - Initialize w_1 and Learning rate $0 < \eta \leq 1$
  - For $t = 1, \ldots, T$ iterations

$$w_{t+1} = w_t - \eta g_t \tag{2}$$

    where $g_t = \nabla f(w_t)$

- Think of g_t as the update vector at time t

Gradient Descent: Intuition



- Intuition: Gradually follow the direction of decrease
- If $f(x) \geq f(y)$, by Taylor's theorem, we have locally

$$\begin{aligned} f(y) &\approx f(x) + \langle y - x, \nabla f(x) \rangle \\ 0 \geq f(y) - f(x) &\approx \langle y - x, \nabla f(x) \rangle \\ 0 &\geq \langle y - x, \nabla f(x) \rangle \end{aligned} \tag{3}$$

That is, gradient locally gives us the direction of increase.

How good is GD on Convex Functions



- Recall by convexity we can bound the suboptimality of a solution w_t, i.e. $f(w_t) - f(w^*)$

$$f(w^*) \geq f(w_t) + \langle w^* - w_t, \nabla f(w_t) \rangle \tag{4}$$

or equivalently

$$f(w_t) - f(w^*) \leq \langle \nabla f(w_t), w_t - w^* \rangle \tag{5}$$

- Thus, it suffices to bound the above inner-product

Proposition

For any sequence of update vectors $g_1, \ldots, g_T \in \mathbb{R}^d$ and any vector $w^* \in \mathbb{R}^d$, the update rule $w_{t+1} = w_t - \eta g_t$ satisfies

$$\sum_{t=1}^{T} \langle g_t, w_t - w^* \rangle \leq \frac{1}{2\eta} \|w_1 - w^*\|^2 + \frac{\eta}{2} \sum_{t=1}^{T} \|g_t\|^2 \tag{6}$$

Proof of the Proposition

- Let us consider the evolution of $\|w_t - w^*\|^2$ and leverage the form of the update

$$\begin{aligned} \|w_{t+1} - w^*\|^2 &= \|w_t - \eta g_t - w^*\|^2 \\ &= \|w_t - w^*\|^2 + 2\eta\langle g_t, w_t - w^*\rangle + \eta^2\|g_t\|^2 \end{aligned} \tag{7}$$

- Rearrange

$$\langle g_t, w_t - w^*\rangle = \frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta} + \frac{\eta}{2}\|g_t\|^2 \tag{8}$$

- Summing over t

$$\sum_{t=1}^{T}\langle g_t, w_t - w^*\rangle = \sum_{t=1}^{T}\frac{\|w_t - w^*\|^2 - \|w_{t+1} - w^*\|^2}{2\eta} + \sum_{t=1}^{T}\frac{\eta}{2}\|g_t\|^2 \tag{9}$$

How good is GD on Convex Functions (Cont'd)



- Recall in GD: $w_{t+1} = w_t - \eta \nabla f(w_t)$ (that is, $g_t = \nabla f(w_t)$)
- By convexity $f(w_t) - f(w^*) \leq \langle \nabla f(w_t), w_t - w^* \rangle$
- Thus, by our proposition

$$\text{Accumulated suboptimality} = \sum_{t=1}^{T} f(w_t) - f(w^*) \leq \frac{1}{2\eta} \|w_1 - w^*\|^2 + \frac{\eta}{2} \sum_{t=1}^{T} \|\nabla f(w_t)\|^2 \quad (10)$$



Definition

A function $f : \mathcal{X} \to \mathbb{R}$ is G-Lipschitz if and only if

$$|f(x) - f(y)| \leq G\|x - y\| \tag{11}$$

for some $G \geq 0$ and all $x, y \in \mathcal{X}$.

- Intuition: Bounded sensitivity and a measure of continuity of a function
- By rearranging we see the slope is bounded

$$\frac{|f(x) - f(y)|}{\|x - y\|} \leq G \tag{12}$$

- Thus, if the function is differentiable, equivalently a function is G-Lipschitz iff $\|\nabla f(x)\| \leq G$

G-Lipschitz Functions: Examples



- $f(x) = \sqrt{x}$
  - $x \in (0, 1]$
  - $x \in [0.5, 1]$
- $f(x) = |x|$
  - $x \in \mathbb{R}$
- $f(x) = e^x$
  - $x \in \mathbb{R}$
  - $x \in [a, b]$
- $f(x) = x^2$
  - $x \in \mathbb{R}$
  - $x \in [1, 10]$

Implications for GD



- Assume f is G-Lipschitz, i.e., $\|\nabla f(x)\| \leq G$
- Let $D = \|w_1 - w^*\|$
- From our proposition

$$\sum_{t=1}^{T} f(w_t) - f(w^*) \leq \frac{1}{2\eta}\|w_1 - w^*\|^2 + \frac{\eta}{2}\sum_{t=1}^{T}\|\nabla f(w_t)\|^2 \tag{13}$$

Theorem

Assume f is G-Lipschitz and convex. GD with the update rule $w_{t+1} = w_t - \eta\nabla f(w_t)$ and learning rate $\eta = \frac{D}{G\sqrt{T}}$ satisfies the following bound on the incurred average suboptimality

$$\frac{1}{T}\sum_{t=1}^{T} f(w_t) - f(w^*) \leq \frac{DG}{\sqrt{T}} \tag{14}$$

How to Output a Solution?

- In practice we output the last solution, i.e. $\hat{w} = w_T$ as it tends to have a better performance
  - We need a different analysis or more structures to prove its performance
- Alternatively, let $\hat{w} = \frac{1}{T}\sum_{t=1}^{T} w_t$
  - By convexity

$$f(\hat{w}) - f(w^*) \leq f(\frac{1}{T}\sum_{t=1}^{T} w_t) - f(w^*) \leq \frac{1}{T}\sum_{t=1}^{T} f(w_t) - f(w^*) \leq \frac{DG}{\sqrt{T}} \tag{15}$$

- Alternatively, let $\hat{w} \sim \mathcal{U}\{w_1, \ldots, w_T\}$
  - On expectation, this randomized solution satisfies

$$\mathbb{E}_{\hat{w}}[f(\hat{w}) - f(w^*)] = \frac{1}{T}\sum_{t=1}^{T} f(w_t) - f(w^*) \leq \frac{DG}{\sqrt{T}} \tag{16}$$

Analyzing the Bound



Theorem

GD with the update rule $w_{t+1} = w_t - \eta \nabla f(w_t)$ and learning rate $\eta = \frac{D}{G\sqrt{T}}$ satisfies

$$\text{Average Suboptimality} \leq \frac{DG}{\sqrt{T}} = \mathcal{O}(1/\sqrt{T}) \tag{17}$$

- The regulating role played by the learning rate, η
- Recall $w \in \mathbb{R}^d$, so no explicit dependence on the intrinsic dimension
- d, however finds its way in $\|\nabla f(x)\| \leq G$ and $D = \|w_1 - w^*\|$, typically
- Iteration Complexity: Number of iterations T to find an ϵ-accurate solution

$$T \geq \frac{D^2 G^2}{\epsilon} = \Omega(1/\epsilon^2) \Rightarrow \text{Average Suboptimality} \leq \epsilon \tag{18}$$

- Can we do better?